

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000789		
	Version:	8.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	19-Dec-2014
Title:	SM DG57. Error code list			

DG-57 Service Manual

ERROR CODES LIST

Procedure description
This document contains a list of all errors reported by the system Note: Document REGD-0000789_v8 replaces document REGD-0000789_v7.

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1. Errors Displayed to the user

Error code	Description	Operator's message	Manual replacement	Automatic Message	Possible cause	Recommended action
CarrErrCod C	The card has been reset	Electronic Hardware failure	N/A	N/A	8V Power supply failure. PCA Failure.	Check 8V Power supply (see document REGD-0000792). Check PCA status.
CarrErrCod D	Time-out positioning motor Y	Could not control sample starting position	For some reason the movement could not be done. Open door, examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
CarrErrCod E	Time-out positioning motor X	Could not control reagent station position	For some reason the movement could not be done. Open door, examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
CarrErrCod F	Current limit of the motor Y	Could not control sample starting position	For some reason the movement could not be done. Open door, examine and retry.	N/A	Sample carrousel blocked. Current limit misadjusted. Encoder cable broken.	Check carousel condition. Check/adjust current limit (REGD-0000794). Check encoder cables.
CarrErrCod G	Current limit in both motors	Could not control reagent station position	For some reason the movement could not be done. Open door, examine and retry.	N/A	Both carrousels blocked. Current limits misadjusted. Encoder cables broken.	Check carousel condition. Check/adjust current limit (REGD-0000794). Check encoder cables.
		Could not control sample starting position	For some reason the movement could not be done. Open door, examine and retry.	N/A		
CarrErrCod H	Current limit of the motor X	Could not control reagent station position	For some reason the movement could not be done. Open door, examine and retry.	N/A	Reagents carrousel blocked. Current limit misadjusted. Encoder cable broken.	Check carousel condition. Check/adjust current limit (REGD-0000794). Check encoder cables.
CarrErrCod K	EEPROM error	Electronic hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part (REGD-0000825).

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CarrErrCod P	Stability failure in motor X	Could not control reagent station position	For some reason the movement could not be done. Open door, examine and retry.	N/A	Faulty motor or PCA.	Replace Motor or PCA.
CarrErrCod Q	Stability failure in motor Y	Could not control sample starting position	For some reason the movement could not be done. Open door, examine and retry.	N/A	Faulty motor or PCA.	Replace Motor or PCA (REGD-0000825).
CarrErrCod R	Fatal error	Electronic hardware failure	N/A	N/A	Jitter in microcontroller's clock line. Bad microcontroller's pins welding. PCA failure.	Restart the system. Replace faulty part (REGD-0000825).
CentErrCod C	The card has been reset	Electronic hardware failure	N/A	N/A	8V Power supply failure. PCA Failure.	Check 8V Power supply (see document REGD-0000792). Check PCA status.
CentErrCod E	Time-out positioning motor X	Impossible to control centrifuge position.	For some reason the movement could not be done. Open door, examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure.	Check for 24V Power supply at PCA and motor. Check encoders sensor performance.
CentErrCod H	Current limit of the motor X	Impossible to control centrifuge position	For some reason the movement could not be done. Open door, examine and retry.	N/A	Centrifuge blocked. Current limit misadjusted. Encoder cable broken.	Check centrifuge condition. Check/adjust current limit (REGD-0000802). Check encoder cables.
		Centrifuge malfunction!	N/A	N/A		
CentErrCod K	EEPROM error	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part (REGD-0000825).
CentErrCod M	Home of motor X is not performed	Impossible to control centrifuge position.	For some reason the movement could not be done. Open door, examine and retry.	N/A	Encoder (home sensor) cable faulty. PCA failure. 8V Power supply failure.	Check cable. Check PCA status. Check 8V Power supply (see document REGD-0000792).
CentErrCod N	Unbalancing sensor activated	Centrifuge malfunction!		N/A	Sensor misadjustment.	Adjust sensor (REGD-0000803).
CentErrCod P	Stability failure in motor X	Impossible to control centrifuge position.	For some reason the movement could not be done. Open door, examine and retry.	N/A	Faulty motor or PCA.	Replace Motor (REGD-0000824) or PCA (REGD-0000825).
		Centrifuge malfunction!		N/A		

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CentErrCod Z	The card has been reset.	Electronic Hardware failure	N/A	N/A	8V Power supply failure. PCA Failure.	Check 8V Power supply (see document REGD-0000792). Check PCA status.
ComErrCod 1	Communication port doesn't answer	Lost communication	N/A	N/A	COM port in computer is already in usage by another program.	Check PC configuration.
ComErrCod 3	A checksum error was detected	Lost communication	N/A	N/A	Cable failure, noise in the line.	Check connections PC-Instrument.
ComErrCod 4	XYZ card doesn't answer	Lost communication	N/A	N/A	PCA failure. 8V failure. Cable failure. RS-232 to RS485 converter failure.	Check PCA status. Check 8V Power supply. Check communications cables (both external and internal). Check converter PCA.
ComErrCod 5	Carrousel card doesn't answer	Lost communication	N/A	N/A	PCA failure. 8V failure. Cable failure. RS-232 to RS485 converter failure.	Check PCA status. Check 8V Power supply. Check communications cables (both external and internal). Check converter PCA.
ComErrCod 6	Centrifuge card doesn't answer	Lost communication	N/A	N/A	PCA failure. 8V failure. Cable failure. RS-232 to RS485 converter failure.	Check PCA status. Check 8V Power supply. Check communications cables (both external and internal). Check converter PCA.
ComErrCod 7	IO card doesn't answer	Lost communication	N/A	N/A	PCA failure. 8V failure. Cable failure. RS-232 to RS485 converter failure.	Check PCA status. Check 8V Power supply. Check communications cables (both external and internal). Check converter PCA.
ComErrCod 9	Wrong command parameters	Lost communication	N/A	N/A	Cable failure, noise in the line.	Check connections PC-Instrument.
IOErrCod 1	Microcontroller watchdog was activated	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part (REGD-0000825).
IOErrCod 2	Microcontroller watchdog was not activated	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part (REGD-0000825).
IOErrCod 4	The card has been reset	Electronic Hardware failure	N/A	N/A	8V Power supply failure. PCA Failure.	Check 8V Power supply (see document REGD-0000792). Check PCA status.

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IOErrCod 6	SPI time-out	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part (REGD-0000825).
IOErrCod 9	EEPROM error	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part (REGD-0000825).
LecErrCod 12	Lack of light homogeneity	Can't read the card	N/A	N/A	Backlight LED burnt.	Replace LED.
LecErrCod 13	Image brightness saturation	Can't read the card	N/A	N/A	Camera misadjustment	Readjust camera (REGD-0000810).
LecErrCod 31	Tube detection impossible	Could not read the tube	N/A	N/A	Camera misadjustment.	Readjust camera (REGD-0000810).
LecErrCod 32	Tube detection impossible	Could not read the tube	N/A	N/A	Camera misadjustment.	Readjust camera (REGD-0000810).
LecErrCod 34	Tube detection impossible	Could not read the tube	N/A	N/A	Camera misadjustment.	Readjust camera (REGD-0000810).
LecErrCod 35	Tube detection impossible	Could not read the tube	N/A	N/A	Camera misadjustment.	Readjust camera (REGD-0000810).
MaqErrCod 10	Insufficient vacuum	Insufficient pressure and/or vacuum	Revise all bottle's caps and wait until this error disappear. If doesn't disappear in less than 1 minute, call technical service.	N/A	Refer to "lack of vacuum / pressure" in document REGD-0000788.	
MaqErrCod 106	Level detection fault	Level not detected in %s.\Retry?	N/A	The Sample will be cancelled after %d seconds	Refer to level detection problems in document REGD-0000788.	
		Card not found. \ Retry?	N/A	N/A		

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Error code	Description	Operator's message	Manual replacement	Automatic Message	Possible cause	Recommended action
MaqErrCod 107	Card is missing	Sample %d not found!\nRetry?	N/A	The Sample will be cancelled after %d seconds	Gripper misadjustment. Gripper cable failure.	Adjust gripper (REGD-0000807, REGD-0000808). Replace gripper cable (REGD-0000823).
		Insufficient Diluent at %d!\nRetry?	Replace Reagent R%d	N/A		
		Level not detected in %s!\nRetry?	Replace Reagent R%d	N/A		
		Unable to pick up card %s!\nRetry?	Remove the closest card from the centrifuge and place it at the second balancing coordinate.	N/A		
		Unable to pick up card %s!\nRetry?	The card's seal seems to be defective. Try to fix and leave it at the second balancing coordinate.	N/A		
		Card lost going to %s!\nRetry?	Place the card at the second balancing coordinate to retry.	N/A		
MaqErrCod 108	Card is present	Unable to pick up card %s!\nRetry?	Remove the closest card from the centrifuge and place it at the second balancing coordinate.	N/A	Gripper malfunction.	Check/replace gripper.
		Unable to pick up card %s!\nRetry?	The card's seal seems to be defective. Try to fix and leave it at the second balancing coordinate.	N/A		
		Unable to release card %s!\nRetry?	The card's seal seems to be defective. Try to fix and leave it at the second balancing coordinate.	N/A		

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Error code	Description	Operator's message	Manual replacement	Automatic Message	Possible cause	Recommended action
MaqErrCod 109	Card not found	Card %s not found.\Retry?	The card's seal seems to be defective. Try to fix and leave it at the second balancing coordinate.	N/A	Gripper adjustment. Gripper cable defective.	Readjust gripper (REGD-0000807, REGD-0000808). Replace cable (REGD-0000823).
MaqErrCod 111	Dilutions well is obstructed	Clot detected in sample %d	N/A	The sample will be cancelled after %d seconds	Clot in sample.	Change sample.
MaqErrCod 114	Solution A fluidic circuit is obstructed. Liquid doesn't come out from probe or cannot be emptied from dilution cup.	Possible obstruction in fluidic system!	Revise all bottle's caps and racors. In case problem persist, call technical service.	N/A	Incorrect bottles connections. Bottle's cap not tightened. Tubes or electro valves obstruction.	Check bottles caps and connections. Replace obstructed tubes or electro valves.
MaqErrCod 115	Solution B fluidic circuit is obstructed. Wash station vacuum is not sufficient and liquid drips during probe wash.	Possible pbstruction in fluidic system!	Revise all bottle's caps and racors. In case problem persist, call technical service.	N/A	Incorrect bottles connections. Bottle's cap not tightened. Tubes or electro valves obstruction.	Check bottles caps and connections. Replace obstructed tubes or electro valves. Clean wash station from possible clots.
MaqErrCod 19	Time-out to positioning centrifuge.	Electronic Hardware failure	N/A	N/A	24V Power Supply failure. Motor failure. Encoder failure.	Check for 24V Power supply at PCA and motor. Check encoder sensor performance.
MaqErrCod 2	Power up failure	There are cards in the service holder	Remove ALL the cards from the service holder	N/A	Cards in service rack.	Remove cards in service rack.
		There were cards in the centrifuge	Remove ALL the cards from the service holder	N/A	Cards in the centrifuge.	Remove cards from centrifuge.
		Card %s not found.\Retry?	Place balancing card in it's position	N/A	No balancing card.	Place a balancing card in position 1.
MaqErrCod 20	Time-out centrifuging	Electronic Hardware failure	N/A	N/A	24V Power Supply failure. Motor failure. Encoder failure.	Check for 24V Power supply at PCA and motor. Check encoders sensor performance.
MaqErrCod 211	Reagent eliminated	Reagent R%d removed or incorrect	Place the reagent AT THE SAME position	N/A	Reagent not present.	Place original reagent in position.
MaqErrCod 3	Digital pump time-out	Electronic Hardware failure	N/A	N/A	24V Power supply failure. Digital pump failure.	Check 24V power supply. Replace pump (REGD-0000831).

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MaqErrCod 318	Time-out to obtain temperature	The incubator does not work correctly	N/A	N/A	Heater/cooler failure. Sensor misadjustment.	Replace Heater PCA/Peltier cells (REGD-0000843). Adjust temperature (REGD-0000799).
MaqErrCod 319	Temperature out of range	The incubator does not work correctly	N/A	N/A	Sensor broken.	Replace Heater PCA.
MaqErrCod 320	Over-temperature	The incubator does not work correctly	N/A	N/A	Sensor broken.	Replace Heater PCA.
MaqErrCod 323	Clot sensor wrong response.	Clot sensor doesn't work.	N/A	N/A	Sensor obstructed. Teflon tube disconnected.	Check Teflon tube, clot sensor or clot PCA. Replace clot PCA.
MaqErrCod 324	Turning off the pumps, pressure has not lowered within 5 seconds.	Possible obstruction in fluidic system!	N/A	N/A	Air filter obstructed. Electro valve D is blocked.	Check air filter and electro valve C.
MaqErrCod 4	Teflon tube obstruction between syringe and probe.	Obstruction in Teflon tube.	N/A	N/A	Obstruction caused by salt contained in saline solution, when instrument has been inactive for long time without performing the final wash.	Repeat priming. Change Teflon tube.
MaqErrCod 424	Wrong velocity during centrifuging	Centrifuge malfunction!	N/A	N/A	24V Power Supply failure. Motor failure. Encoder failure.	Check for 24V Power supply at PCA and motor. Check encoders sensor performance.
MaqErrCod 5	Gripper blocked	Unable to release the card	Remove the card from the gripper by pushing the gripper shaft.	N/A	Gripper malfunction.	Check/replace gripper.
MaqErrCod 7	Reagents carousel time-out	Electronic Hardware failure	N/A	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
MaqErrCod 9	Insufficient pressure	Insufficient pressure and/or vacuum	Revise all bottle's caps and wait until this error disappear. If doesn't disappear in less than 1 minute, call technical service.	N/A	Refer to "lack of vacuum / pressure" in document REGD-0000788.	

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Error code	Description	Operator's message	Manual replacement	Automatic Message	Possible cause	Recommended action
XerErrCod 1	Error during initialization	Syringe is not working	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Syringe not working. 24V Power supply failure.	Check/replace syringe (REGD-0000831). Check 24V Power supply.
XerErrCod 10	Current limit digital pump valve	Syringe is not working	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 11	Not allowed movement	Lost communication	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 15	Overflow	Lost communication	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 2	Wrong command	Lost communication	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 20	Communication error	Lost communication	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 3	Wrong operand	Lost communication	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 4	Wrong sequence	Lost communication	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 6	EEPROM error	Electronic Hardware failure	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 7	Digital pump not initialized	Electronic Hardware failure	N/A	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XerErrCod 9	Current limit digital pump	Syringe is not working	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Syringe not working.	Check/replace syringe (REGD-0000831).
XYZErrCod d	Time-out positioning of the gripper motor.	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
XYZErrCod g	Current limit of the gripper motor	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Gripper Motor blocked. Current limit misadjusted. Encoder cable broken.	Check Motor condition. Check/adjust current limit (REGD-0000806). Check encoder cables.
XYZErrCod 1	Microcontroller watchdog was activated	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part REGD-0000825).

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XYZErrCod 2	Microcontroller watchdog was not activated	Electronic Hardware failure	N/A	N/A	PCA/EEPROM failure.	Replace faulty part REGD-0000825).
XYZErrCod 4	The card has been reset	Electronic Hardware failure	N/A	N/A	8V Power supply failure. PCA Failure.	Check 8V Power supply (see document REGD-0000792). Check PCA status.
XYZErrCod B	Time-out positioning motor X	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
XYZErrCod C	Time-out positioning motor Y	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
XYZErrCod D	Time-out positioning probe motor	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	24V Power Supply failure. Motor failure. Encoder failure. Home sensor failure.	Check for 24V Power supply at PCA and motor. Check encoders and home sensor performance.
XYZErrCod E	Current limit of the motor X	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	X Motor blocked. Current limit misadjusted. Encoder cable broken.	Check Motor condition. Check/adjust current limit (REGD-0000806). Check encoder cables.
XYZErrCod F	Current limit of the motor Y	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Y Motor blocked. Current limit misadjusted. Encoder cable broken.	Check Motor condition. Check/adjust current limit (REGD-0000806). Check encoder cables.
XYZErrCod G	Current limit of the probe motor	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Probe Motor blocked. Current limit misadjusted. Encoder cable broken.	Check Motor condition. Check/adjust current limit (REGD-0000806). Check encoder cables.
XYZErrCod I (n)	Probe crashed against some obstacle (impact). Code error is followed by a number 'n' which means the number of impacts that the probe has suffered since last software reset.	XYZ movement Impossible	For some reason the movement could not be done. Examine and retry.	N/A	Sample tubes or reagent bottles plugged. Probe positioning misadjustment.	Check sample tubes or reagent bottles. Adjust probe positioning.

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XYZErrCod K	Tolerance error in positioning the motor X	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Faulty motor or PCA.	Replace Motor or PCA (REGD-0000825).
XYZErrCod L	Tolerance error in positioning the motor Y	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Faulty motor or PCA.	Replace Motor or PCA (REGD-0000825).
XYZErrCod M	Tolerance error in positioning the probe motor	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Faulty motor or PCA.	Replace Motor or PCA (REGD-0000825).
XYZErrCod O	Tolerance error in positioning the gripper motor	XYZ movement Impossible	For some reason the movement could not be done. Open door, Examine and retry.	N/A	Faulty motor or PCA.	Replace Motor or PCA (REGD-0000825).
XYZErrCod Q	Home error in X axe	Electronic Hardware failure	N/A	N/A	Home sensor faulty.	Check/replace home sensor.
XYZErrCod R	Home error in Y axe	Electronic Hardware failure	N/A	N/A	Home sensor faulty.	Check/replace home sensor.
XYZErrCod S	Home error in probe axe	Electronic Hardware failure	N/A	N/A	Home sensor faulty.	Check/replace home sensor.
XYZErrCod U	Home error in gripper axe	Electronic Hardware failure	N/A	N/A	Home sensor faulty.	Check/replace home sensor.
XYZErrCod V	Home not performed	Electronic Hardware failure	N/A	N/A	Home sensor faulty.	Check/replace home sensor.
XYZErrCod Y	Level detection PCA does not respond.	Electronic Hardware failure.	N/A	N/A	Level detection PCA malfunctioning. Cable cut between XYZ and level detections PCA.	Check Level detection PCA and communication cable between it and XYZ PCA.

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2. Errors showed in TS software or logged in the log file.

Error code	Description	Possible cause	Recommended action
CarrErrCod I	Wrong arguments	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000797)).
CarrErrCod A	Wrong parameters	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000797)).
CarrErrCod J	Parity error	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000797)).
CarrErrCod L	Communications time-out	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000797)).
CarrErrCod N	The door is open	The General door is open.	Close the general door, check door latch.

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CarrErrCod M	Home of motor X is not performed	Samples Carrousel home is not performed (position not initialized).	Perform a Home of Samples Carousel to initialize it.
CarrErrCod O	Time-out of the gripper	Gripper did not reach programmed position within expected time.	Check for 24V Power supply at PCA and motor and motor cable. Check gripper adjustments (REGD-0000807 and REGD-0000808).
CarrErrCod S	Noise in communications	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000797)).
CarrErrCod T	Home of motor Y is not performed	Reagents Carrousel home is not performed (position not initialized).	Perform a Home of Samples Carousel to initialize it.
CarrErrCod Z	Fatal error	General error in PCA.	Check PCA (REGD-0000797).

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Error code	Description	Possible cause	Recommended action
CarrErrCod B	Unknown command	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000797)).
CentErrCod J	Parity error	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000804)).
CentErrCod L	Communications time-out	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000804)).
CentErrCod B	Unknown command	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000804)).
CentErrCod S	Noise in communications	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000804)).

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Error code	Description	Possible cause	Recommended action
CentErrCod I	Wrong arguments	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000804)).
CentErrCod R	Fatal Error	Jitter in microcontroller's clock line. Bad microcontroller's pins welding. PCA failure.	Restart the system. Replace faulty part (REGD-0000825).
ComErrCod 0	Undefined communications error	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA).
ComErrCod 4	XYZ PCA communications error	Communication cable faulty or misconnected. XYZ PCA faulty.	Check communication with XYZ PCA (REGD-0000805).
ComErrCod 5	Carrousel PCA communications error	Communication cable faulty or misconnected. Carrousel PCA faulty.	Check communication with Carrousel PCA (REGD-0000797).

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Error code	Description	Possible cause	Recommended action
ComErrCod 6	Centrifuge PCA communications error	Communication cable faulty or misconnected. Centrifuge PCA faulty.	Check communication with Centrifuge PCA (REGD-0000804).
ComErrCod 7	I/O PCA communications error	Communication cable faulty or misconnected. I/O PCA faulty.	Check communication with I/O PCA (REGD-0000821).
IOErrCod N	Communications error	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000821)).
IOErrCodP	Parity error	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000821)).
IOErrCod J	Wrong arguments	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000821)).

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IOErrCod T	Communications time-out	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000821)).
LecErrCod 9	Reader detects a badly positioned card	Barcode or wells are not detected correctly.	Check reader camera adjustment (REGD-0000812).
LecErrCod 11	Vial not found.	No Reagent vial is found.	Check presence of vial, check reagents camera adjustment (REGD-0000811).
LecErrCod 2	No bar code detected	Barcode cannot be read.	Check barcode quality, check cameras adjustment (REGD-0000810, REGD-0000811, REGD-0000812).
LecErrCod 4	Error in calculating tube width	Samples tube width cannot be calculated.	Check that the label is correctly read, check samples camera adjustment (REGD-0000810).

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Error code	Description	Possible cause	Recommended action
LecErrCod 5	Error during reader calibration procedure	Error occurred when generating the reference image.	Check reader camera adjustment (REGD-0000812).
LecErrCod 6	Wrong type of bar code	Type of barcode detected is not acceptable.	Check barcode type.
LecErrCod 7	Bar code not defined	Barcode does not correspond to any know reagent.	Check barcode.
LecErrCod 8	Reader doesn't detect any card	No card is detected when trying to read.	Check that the card was not lost during transport. Check camera adjustment (REGD-0000812).
LecErrCod 10	Error in the Suport.def file.	Errors in supports.def file.	Check that supports.def file corresponds to the latest version and check its integrity.

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Error code	Description	Possible cause	Recommended action
LecErrCod 1	Bar code unreadable	Couldn't read the barcode.	Check barcode quality. Check camera adjustment (REGD-0000810, REGD-0000811, REGD-0000812).
MaqErrCod 209	Sample eliminated	Sample was removed from the instrument.	Place again sample in the carousel.
MaqErrCod 4	Clot	Clot in sample.	Change sample.
MaqErrCod 8	Samples carousel time-out	Samples carousel did not reach programmed position within expected time.	Check samples carousel module (REGD-0000797).
MaqErrCod 210	Hardware failure of the carousels system.	Hardware failure in carousels PCA.	Check carousel module (REGD-0000797).

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Error code	Description	Possible cause	Recommended action
MaqErrCod 6	Illegal carousel movement	Carousel was trying to move while the probe isn't at home position. Only in Technical Service Software.	Ensure to move home the probe before moving the carousel.
MaqErrCod 423	User hazard caused by the centrifuge	The door was opened while centrifuging and the arm was not placed on top of the centrifuge. It happens if the user opens the door with the security cord.	Train the user not to use the security cord to open the door.
MaqErrCod 18	Out of range in centrifuge positioning.	Centrifuge did not position correctly.	Centrifuge will reposition automatically.
MaqErrCod 11	Washing solutions/waste bottles incidence.	Wash solution bottles empty or Waste bottle full.	Check bottles levels. Check load cells (REGD-0000819, REGD-0000821).
MaqErrCod 422	Hardware failure of the centrifuge system.	Centrifuge PCA is damaged.	Check Centrifuge PCA (REGD-0000804).

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Error code	Description	Possible cause	Recommended action
MaqErrCod 17	Tolerance error in centrifuge positioning.	Centrifuge did not position in the expected position within expected range.	Centrifuge will retry automatically.
MaqErrCod 16	Illegal centrifuge movement	The centrifuge tried to move while the gripper is lowered. Only in Technical Service Software.	Ensure to home the gripper before moving the centrifuge.
MaqErrCod 14	Communication error with the IO card.	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000821)).
MaqErrCod 12	Time-out to obtain vacuum	Vacuum is not reached within expected time.	Check the fluid system (REGD-0000821).
MaqErrCod 13	Time-out on valve operation	Communication to Digital Pump faulty. Digital Pump faulty.	Check Digital pump and communication.

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Error code	Description	Possible cause	Recommended action
MaqErrCod 317	Hardware failure of the IO system.	I/O PCA faulty.	Check I/O PCA (REGD-0000821).
XYZErrCod a	Card already detected	Gripper malfunction.	Check/replace gripper.
XYZErrCod b	Card not found	Gripper adjustment. Gripper cable defective.	Readjust gripper (REGD-0000807, REGD-0000808). Replace cable (REGD-0000823).
XYZErrCod h	No level detected	Refer to level detection problems in document REGD-0000788.	Refer to level detection problems in document REGD-0000788.
XYZErrCod l	Insufficient level detected	Refer to level detection problems in document REGD-0000788.	Refer to level detection problems in document REGD-0000788.

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Error code	Description	Possible cause	Recommended action
XYZErrCod m	Tolerance error in level detection	Refer to level detection problems in document REGD-0000788.	Refer to level detection problems in document REGD-0000788.
XYZErrCod n	Incorrect level detection	Refer to level detection problems in document REGD-0000788.	Refer to level detection problems in document REGD-0000788.
XYZErrCod A	Cannot pick up a card.	Gripper misadjustment. Gripper cable failure.	Adjust gripper (REGD-0000807, REGD-0000808). Replace gripper cable (REGD-0000823).
XYZErrCod T	Communication time-out	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000805)).
XYZErrCod P	Parity error	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000805)).

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Error code	Description	Possible cause	Recommended action
XYZErrCod N	Communication Noise	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000805)).
XYZErrCod 8	EEPROM error	Error in EEPROM.	Replace EEPROM and readjust and check Robot module (REGD-0000805, REGD-0000806, REGD-0000807, REGD-0000808).
XYZErrCod 9	EEPROM error	Error in EEPROM.	Replace EEPROM and readjust and check Robot module (REGD-0000805, REGD-0000806, REGD-0000807, REGD-0000808).
XYZErrCod J	Wrong command parameters	Communication cable faulty or misconnected. RS-232 to RS-485 PCA faulty. Affected PCA faulty.	Check communication between PC and instrument (cable, RS-232 PCA and affected PCA (REGD-0000805)).

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3. Error Incidents

Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
Barcode not found	The bar code ID entered is not contained in any batch.	Re-enter the bar code ID or enter a correct bar code ID.
Barcode already present	Repeated sample ID	- Verify the correct tube option is selected - Verify proper sample labeling
Well incident in Card	- Gel card has no available wells for use - Possible air bubbles in the well(s) - Possible scratches on the well (s)	Replace with a new Gel card of the same type
Unable to read barcode	- The sample bar code is not positioned correctly or is in poor condition. - The bar code is too faint. - Barcode not present	- Replace with a suitable bar code label, OR - Manually identify the sample (see "Manually identifying Samples and Assigning Tests").
Cannot read the card	- The card bar code is not positioned correctly or is in poor condition. - The bar code is too faint.	- Turn the card with barcode facing to the left - Replace with a new Gel Card of the same type
Bottle check	Displays during the pre-test self-test check when - the waste bottle is full - the wash solution bottles are empty, or - the bottles are badly positioned	- Check the bottle fluid levels - Verify the bottles are correctly positioned.
Cards pending checking on the service rack	Gel cards to be reviewed not removed from service rack	Remove and review results and take appropriate actions.

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Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
Diameter not detected	- The sample tube bar code label is incorrectly positioned. - The tube is tilted.	- Properly adhere the bar code label to the tube. - Tighten the spring-clips on the sample carousel.
Reagent(s) not present: xxx	- The correct reagent has not been loaded in the reagent carousel. - The bar code has not been read correctly. - The bottle has not turned correctly.	- Load the correct reagent. - Manually identify the reagent; use bottles with bar codes in good condition. - Reposition the bottle;
X of Y type cards are missing	- The number of gel cards of the specified type is insufficient. - The gel card bar code label is in poor condition. - A sufficient number of gel cards of the specified type exist, but one or more have defective or used wells, or insufficient liquid level.	- Remove the problematic gel cards (displayed on the screen in red) - Load more gel cards of the specified type.
Reagent expired	The reagent has passed its expiration date.	Change the reagent for one that is still in date.
RBC and serum barcodes do not agree	The software has detected a sample with two tubes but different IDs.	- Click the red X to interrupt execution - correct the incident, and - restart sample identification.
Tube not detected	- Sample tube not present in the sample carousel. - Sample tube not in the correct position	- Load the sample on the sample carousel. - Correct sample position on the sample carousel.
Sample not found	- The software cannot read the bar code label (label is incorrectly positioned, has come unstuck, etc.). - A position on the sample carousel with no tube present has been marked manually.	- Verify the bar code label is correctly positioned and adheres to the tube. - Click the red X to interrupt execution - correct the incident, and - restart sample identification.

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Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
The batches ### have failed the last control. See the batch listing for more details.	The result for the quality control per-formed on the test is QC Fail.	1. Check all routine laboratory practices and materials used make necessary corrections, and 2. Repeat the test as appropriate. 3. The software never cancels the results of a batch that has failed 4. Quality control. The user is responsible for taking all necessary corrective measures.
Unexpected reagent	When rechecking reagent IDs, the bar code does not match the one previously identified.	Verify the reagent type is correct verify the bar code label is in good condition, and load the reagent properly.
There are identification errors. Abort and correct it.	A previously identified barcode labeled sample tube cannot be properly re-identified by the instrument once the door had been opened during testing.	If the sample tube has been turned around so the barcode label cannot be read, turn the sample tube until the barcode label is visible.
	The sample tube barcode cannot be re-identified either because it has been turned around or has been replaced by a different barcode labeled sample tube.	If the original sample tube has been incorrectly replaced with a different sample tube, determine which is the correct sample to be tested.
Barcode does not agree. Try Again.	While manually entering a sample ID number, the confirmation sample ID number does not match the original sample ID number entry.	Verify the correct sample ID number and enter it as both the original and confirmation numbers.
Error in MdigGrap : Error from Meteor-II low level function.	1. Instrument is turned off while software is started.	1. Turn the instrument on prior to starting the software.
	2. Camera cable has been disconnected.	2. Check camera cable, if disconnected or loose, reconnect and start again.
Click cards on service rack and extract it from the equipment	1. Some wells may be unused	1. Place the card back on the incubator to be used at a later time.
	2. Some results may need to be reviewed.	2. Manually review the results and modify results accordingly.
Card is no longer on the service rack	The card scanned has already been used.	Scan the card to be used

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Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
Error: No batch available. OK	There is no batch data available in the Tandas folder and the user accesses one of the following software features: • The "Batch reading" option of the Results software module. • The "Results by batch" option of the Data Management module. • The "Results by sample" option of the Data Management module. • The "Batch listing" option of the Data Management module. • The "Speed button" option of the Data Management module.	The options should display the expected data once data is available in the Tandas folder
There are some samples to be processed. Deleted them? Yes / No	This is an informational message that indicates there are samples that have been previously defined to be tested will be deleted.	Answer "yes" if the samples are to be deleted. Answer "No" if the samples are not to be deleted.
Impossible to initialize the instrument. OK	Instrument cannot initialize due to mechanical failures (i.e. the probe does not move)	Re-start the instrument and the software.
Tube doesn't exists! Delete it? Yes / No	The sample was manually defined, but during automatic re-identification, the system did not find a tube in that position.	Load the sample in the position where the sample was defined manually.

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4. Informational Incidents

Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
Cancelled samples	Samples were cancelled during test execution due to presence of a clot, repeated ID, insufficient sample level, etc.	- Accept the cancellation. - Repeat the test for the cancelled sample.
Cards are placed in service rack	- The gel cards require a manual check. Any of the following causes may apply: - volume verification detected levels outside the established limits - the final dispensed volume could not be detected - Gel Card placed on service rack based on configuration settings - There are still available wells in the Gel Card - There are questionable results for one or more wells.	- View the results file to determine the entry for the well: ~ = volume detected is outside established limits “agglutination result + !” = final volume not determined - Remove and review results and take appropriate actions. - Remove card and store card per facility SOP for later use. - Remove and review results and take appropriate actions.
There are repeated discordant wells. It's recommended to cancel them to delete the discordance.	By modifying the result for a repeated well (same well appearing in more than one batch for the same sample), the results for the wells in the different batches are contradictory	Either modify the results so that all well results match for the same sample appearing in different batches, or cancel the well. Reference: See “Modifying Test Results” on page 8-28.
There wasn't enough reagent volume. Recheck before begin?	One of the necessary reagents doesn't have sufficient volume	Place another vial with sufficient volume of reagent on the reagent rack.
Card's bar code does not agree	The gel card ID at the end of the run does not match the gel card ID that was identified at the beginning of a run	Repeat testing.

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Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
The last quality control has failed!	The previous quality control for the test being requested failed.	Perform quality control for the test being requested prior to defining the test.
QC samples for test [xxx] not included in this batch [sample ID] Continue anyway?	One or more of the quality control samples for the test to be executed has/have not been identified. Note: Processing the test — <i>without any required quality control samples</i> results in a processing status for the batch of "Read." — <i>with only some of the required quality control samples</i> results in a processing status for the batch of either	Either load the missing quality control samples and restart test execution, or continue execution with some or no quality control samples. (No quality control will be performed.)
There are empty bottles on the carousel	A change of panel has been per-formed due to one of the reagents running out.	Remove the empty bottle.
There are receivers without donors. Continue anyway? (OK/Cancel)	Patient samples assigned to a cross-match test have not been assigned donors.	Take one of the following actions: Display the worksheet (<i>Sheet</i> tab, <i>Execution</i> window) to determine the test(s) to which the donor samples have been assigned. Correct test assignment (using the <i>Test</i> tab) and click the green arrow to continue processing. Click <i>Abort pipetting</i> on the <i>Stop</i> window to cancel processing. Click <i>???</i> to continue processing. (The patient sample will not be processed).
No Level	A sample tube that has been placed in the sample carousel and assigned a test is found to be empty during batch testing.	Either remove the empty sample tube or add sufficient sample volume to the empty tube.

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Informational Message	Possible Cause(s) of Error	Corrective Actions Taken
Some wells must be reviewed	A gel card has been placed in the service rack to be reviewed by the technologist. Gel cards may be placed in the service rack for any of the following reasons: - A reaction result was obtained that had been previously defined in the Configuration screen to be reviewed - A microtube result is a “?” - A microtube result is a “E” - The instrument detected a possible volume issue with one or more wells	Review the reaction results and modify if necessary.
Some wells are unused in card	A gel card was placed in the service rack that contains wells that are still available for future testing.	Remove the gel card from the service rack and properly store per facility SOP's for future use.
The next batches aren't signed: xxxx	This message will appear if batch data stored in the Tandas folder has been altered and the user attempts to access either the Results software module or the Data Management software module.	Click “OK” and the user is granted access to either the Results or Data Management software module.
Empty the container for waste cards.	Warning to inform the user to empty the waste container so it does not become full during batch processing.	Data for the batch number indicated in the error message is not accessible. Empty the waste container and click “OK” on the message window.
OK		
Cancelled while centrifuging	This message indicates that testing was aborted during the centrifugation process.	Click “OK”. Address the issue that caused the centrifugation process to be aborted. Repeat testing.
Aborted Test	If testing is aborted during the pipetting or incubation process, this message appears in the Sample tab field that normally contains the sample ID number. The message appears in the fields of all sample(s) involved in the aborting process.	Click on the green arrow.
Accept to Continue		Address the issue that caused the pipetting or incubation process to be aborted. Repeat testing.
Unsuccessful instrument initialization.	This message will appear if the user selects the “STOP” button on the “Lost Communication” error message. These messages indicate the RS232 cable has become disconnected.	Click “OK”. Reconnect the RS232 cable. Re-start the software.
Remember there are batches pending to send to host.	User is sending bidirectional batches with unidirectional method.	Instruct user to use bidirectional procedure to send batches to host.

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LIS Error Codes

Error	Possible Cause(s) of Error	Corrective Actions Taken
Host file error #1	File communication error: writing error in the exchange file.	Check space availability and user rights in the exchange directory directory.
Host file error #2	File communication error: writing error in the exchange file (In.dat file not found)	Check whether in.dat is present in the exchange directory
Host file error #3	File communication error: exchange file cannot be deleted.	Check in.dat file user rights (they may be read-only). Check that the file is not already open. Check Host program.
Host file error #4	File communication error: exchange file (Out.dat) still exists after trans-mission ended)	The Host program hasn't deleted out.dat file. Check Host program.
Host format error #1	File communication error: Heading error; wrong separators	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #2	File communication error: Heading error; the first line of the message does not begin with "H"	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #3	File communication error: Error reading patient. No patient ID	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #4	File communication error: Test: Not enough fields	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #5	File communication error: Test: No specimen field	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #6	File communication error: Test: No test name fields	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #7	File communication error: Error: An O before a P field	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #8	File communication error: Error: An R field before a P field	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #9	File communication error: Results: Not enough fields	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host format error #10	File communication error: First line must be the header	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.

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Title:	SM DG57. Error code list			

Error	Possible Cause(s) of Error	Corrective Actions Taken
Host serial error #1	Time-out when receiving frame	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #2	Incorrect frame received	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #3	Incorrect frame received, STX fault	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #4	Neither ETB or ETX being received in the frame	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #5	Frame received without checksum	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #6	Frame received with wrong ck	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #7	Frame received with wrong ID	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #8	Communication error in establishing the connection while awaiting reception.	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #10	Generic communication error	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Host serial error #13	EOT missing	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.
Timeout Exceeded	Timeout configured in the software has passed without having a correct connection. Can appear in junction with other error messages.	Check transmission media (cable connection, interferences). Check Host program message format. Retry communication.

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
4.0	2005-05-06	Updated user displayed error's list and added the deleted ones to the showed in TS software or logged in log file's list, correction of English wording. Update LIS Error Codes list.	Update LIS Error Codes list.
5.0	2008-03-07	Updated list of error displayed to the user and in TS Software and Informational Incidents section with new messages of software version 3.0. Old error codes eliminated.	Update Error Codes list.
6.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY-MM-DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
6.0	2011/04/28	Updated 'XYZerrcod I' description.	'XYZerrcod I' error is always followed by a number.
6.0	2011/04/28	Replaced 3.0 software version by 3.2 software version inside 'Procedure description' section.	Update 'Procedure description' section to 3.2 software version.
6.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market
7.0	July 2011	Document REGD-0000789_v7 replaces document REGD-0000789_v6.	Document version changed.
7.0	July 2011	Document title.	Commercial reference substituted by project reference.
7.0	July 2011	Document header.	Increase information to ease document search.
8.0	December 2014	Corrective actions update	Add / modify some corrective actions considering the feedback received from the field

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